

LONG DURATION TIMER Using Supercapacitor

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When charging devices like mobiles and laptops, it is easy to forget to switch off the power once they are fully charged, often resulting in devices remaining connected to the AC mains supply for hours. A simple solution to this oversight is presented here. The circuit shown in Fig. 1 can automatically switch off any device connected to the AC mains supply after a period ranging from one to over four hours. Extending the delay period is possible by

adjusting the values of the resistors used in the circuit.

Circuit and working

Fig. 2 shows the circuit diagram of a long-duration timer using a supercapacitor. The circuit comprises a centre-tapped transformer (X1), four rectifier diodes (1N4001, D1-D4), a dual op-amp (LM358, IC1), a transistor (2N3904, T1), a 5V SPDT relay (RL1), two LEDs (LED1, LED2), and additional components.

The 230V AC is stepped down

to 6V-0-6V AC by transformer X1, then rectified by diodes D1 and D2 and stabilised by filter capacitor C1, resulting in around 6V DC to power the circuit. The circuit utilises one op-amp (IC1A) of the LM358, while the second op-amp remains unused. A supercapacitor (C2) serves as the heart of the circuit.

Timer circuits can be built by charging a capacitor (C) through a resistor (R). The delay produced by these circuits is calculated as $R \times C$. When the voltage across capacitor C reaches a certain threshold, the circuit triggers, switching the connected device on or off. However, conventional timers using standard capacitors are limited to short delays. For instance, a $1000\mu\text{F}$ capacitor with a 1-mega-ohm resistor produces a delay of about 1000 seconds, or roughly 16 minutes. To achieve significantly longer durations, supercapacitors, with capacitances in farads, are employed, as opposed to common capacitors with capacitances in micro-farads.

In the circuit shown in Fig. 2, a 1F supercapacitor (C2) charging through a 3.9-kilo-ohm resistor (R2) produces a delay (RC) of 65 minutes. The maximum voltage rating of supercapacitor C2 is 2.7V. LED1 ensures the voltage across C2 does not exceed this limit and that the capacitor charges from a voltage source independent

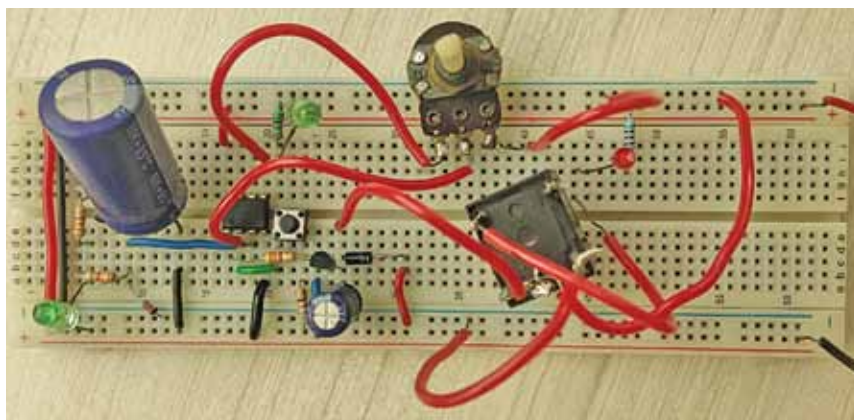


Fig. 1: EFY lab's prototype tested on a breadboard without a transformer

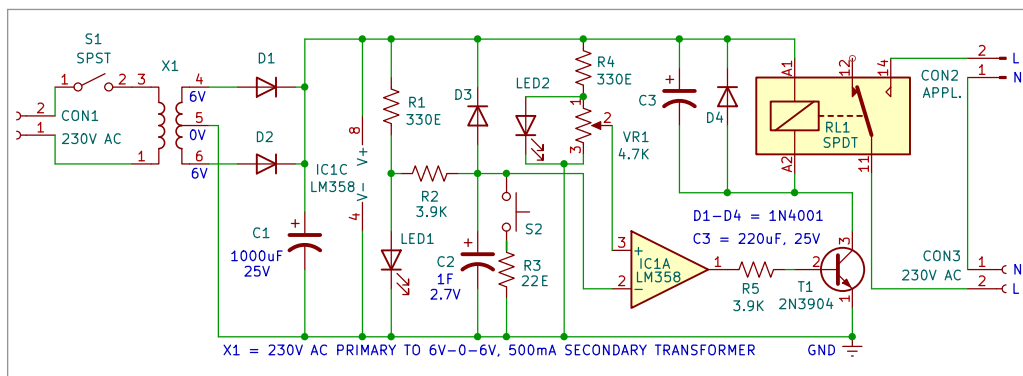


Fig. 2: Circuit diagram